

7th Grade - Science

Photosynthesis

Chapter: Life Processes in Plants

Duration: 45 min

CBSE CURIOSITY

Learning Objectives

- Identify the four essential requirements for photosynthesis: water, sunlight, carbon dioxide, and chlorophyll.
- Explain the process of photosynthesis and write its word equation.
- Describe the conversion of glucose into starch for storage in plants.
- Recognize the contributions of Indian scientist Rustom Hormusji Dastur to the study of photosynthesis.

Previous Knowledge

- Students should know that plants are living things that require water and sunlight to grow, and that most plants have green leaves.

Key Textbook Points

- Photosynthesis components
- Word equation of photosynthesis
- Glucose vs Starch
- Scientific contribution of R.H. Dastur

Teaching-Learning Materials

- Science textbook (Curiosity)
- Blackboard and chalk/markers
- Chart showing Fig. 10.5 (Photosynthesis diagram)
- Flashcards with labels: Carbon Dioxide, Water, Sunlight, Chlorophyll, Glucose, Oxygen

Introduction / Warm-up

- Start by asking students: 'Where do we get our energy from?' After they answer 'food', ask: 'But where do plants get their food? Do they eat like us?' Introduce the concept that plants are unique because they produce their own food through a process called photosynthesis, using simple ingredients from their surroundings.

Explanation of Concept

- Photosynthesis is the process by which green plants prepare their own food. Using the textbook context, explain that plants take in carbon dioxide from the air and water from the soil. In the presence of sunlight and a green pigment called chlorophyll, these ingredients are transformed into carbohydrates. The primary food produced is glucose, a simple carbohydrate that provides instant energy. Any extra glucose is converted into starch for long-term storage. Oxygen is released as a byproduct during this process.

Lesson Flow

1. 0-5 | Introduction

Teacher: Ask brainstorming questions about plant nutrition and introduce the term 'Photosynthesis'.

Students: Participate in discussion and share prior knowledge about plants needing sunlight.

2. 5-15 | Explanation

Teacher: Explain the process using Fig 10.5 and write the word equation on the board.

Students: Listen and note down the word equation in their notebooks.

3. 15-25 | Activity

Teacher: Organize the 'Equation Role Play' activity.

Students: Act out the roles of different components (Sunlight, CO₂, etc.) to form the equation.

4. 25-35 | Direct Instruction

Teacher: Introduce Scientist Rustom Hormusji Dastur and his findings on water and temperature.

Students: Read the 'Know a Scientist' section and discuss how temperature might affect plants.

5. 35-40 | Assessment

Teacher: Ask oral check-for-understanding questions.

Students: Answer questions based on the textbook text.

6. 40-45 | Closure

Teacher: Summarize the key points and assign homework.

Students: Pack up and note down the homework assignment.

Classroom Activity

- Equation Role Play: Assign students roles as 'Carbon Dioxide', 'Water', 'Sunlight', 'Chlorophyll', 'Glucose', and 'Oxygen'. Have the 'Ingredients' (CO₂ and Water) stand on the left. 'Sunlight' and 'Chlorophyll' stand in the middle as facilitators. The 'Ingredients' then move across to become 'Glucose' and 'Oxygen' on the right side to demonstrate the chemical transformation.

Main Concept Details

- The chemical process is summarized by the word equation: Carbon dioxide + Water (in the presence of Sunlight and Chlorophyll) yields Glucose + Oxygen. This represents a transformation of raw materials into chemical energy (carbohydrates).

Daily Life Connection

- Glucose provides instant energy for the plant's life processes.
- Starch acts as a stored food reserve, similar to how humans store fat or glycogen.
- Oxygen produced is essential for the survival of most living organisms on Earth.

Differentiation

- Core: Complete the word equation and explain the difference between glucose and starch.
- Support: Provide a fill-in-the-blanks version of the photosynthesis equation and a pre-drawn diagram for labeling.
- Challenge: Research and explain why R.H. Dastur might have found the 'color of light' important for photosynthesis.

Assessment

1. What are the four things necessary for photosynthesis? (1 mark)
2. In what form is food initially produced in plants? (1 mark)
3. What does glucose get converted into for storage? (1 mark)
4. Write the word equation for photosynthesis. (1 mark)
5. Which gas is released during photosynthesis? (1 mark)
6. Who was Rustom Hormusji Dastur and what did he study? (1 mark)
7. Where does the carbon dioxide used by plants come from? (1 mark)

Board Work

- Topic: Photosynthesis
 - Equation: Carbon dioxide + Water $\xrightarrow{\text{Sunlight/Chlorophyll}}$ Glucose + Oxygen
 - Key Terms:
1. **Glucose: Simple carbohydrate (Instant energy)**
 2. **Starch: Stored carbohydrate**
 3. **Chlorophyll: Green pigment**
- Scientist: R.H. Dastur (Studied effects of water, temperature, and light color on photosynthesis)

Homework

1. Draw a neat labeled diagram of photosynthesis.
2. Write a short paragraph on the contributions of Rustom Hormusji Dastur to Botany.

Learning Outcome

- Students will be able to describe the process of photosynthesis, identify its necessary components and products, and explain how plants store energy as starch.

Teacher Notes

- Ensure students understand that chlorophyll and sunlight are catalysts/conditions and not 'ingredients' consumed in the same way as CO₂ and water. Emphasize the Indian scientist's contribution to instill national pride in scientific achievements.